

Experimental Design - Homework 6

Pranav Tikkawar

May 10, 2026

7.5 Consider the data from the first replicate of Problem 6.11. Construct a design with two blocks of eight observations each with $ABCD$ confounded. Analyze the data.

Solution: Using only replicate I of the 2^4 design from Problem 6.11 (16 runs), we form two blocks of size 8 by computing

$$ABCD = ABCD$$

and assigning runs with $ABCD = -1$ to block B1 and runs with $ABCD = +1$ to block B2. Thus the block effect is completely aliased with the four-factor interaction:

$$\text{block} \equiv ABCD.$$

We fit the full 2^4 model with a fixed block effect. With 1 intercept, 1 block effect, and all 15 factorial effects in 16 runs, the model is saturated, so there is no residual error and ANOVA F -tests are not meaningful. From the fitted coefficients, the largest effects (in magnitude) are

$$A \approx -10, \quad D \approx 5, \quad AB \approx 4.5, \quad ABC \approx -6, \quad ABD \approx 4.8,$$

while the remaining main effects and interactions are much smaller.

In summary, blocking on $ABCD$ creates two blocks of eight runs and protects against block-to-block variation at the cost of confounding the $ABCD$ interaction. The key conclusions agree with the original unblocked analysis: factor A strongly reduces yield at its high level, D increases yield, and the interactions AB , ABC , and ABD are also important.

7.6 Repeat Problem 7.5 assuming that four blocks are required. Confound ABD and ABC (and consequently CD) with blocks.

Solution: Again using replicate I from Problem 6.11, we now create four blocks of size 4 by computing

$$ABD = ABD, \quad ABC = ABC, \quad CD = CD,$$

and defining blocks by the signs of (ABD, ABC) :

$$\begin{aligned} \text{B1: } & ABD = -1, \quad ABC = -1, \quad CD = +1, \\ \text{B2: } & ABD = -1, \quad ABC = +1, \quad CD = -1, \\ \text{B3: } & ABD = +1, \quad ABC = -1, \quad CD = -1, \\ \text{B4: } & ABD = +1, \quad ABC = +1, \quad CD = +1. \end{aligned}$$

Within each block, ABD , ABC , and CD are constant, so these three effects are fully confounded with blocks and cannot be estimated separately.

We fit a model with block, all main effects, and the two-factor interactions AB, AC, BC, AD, BD . The ANOVA shows that the block effect is not significant ($p \approx 0.14$). Factor A remains highly significant ($F \approx 20.5$, $p \approx 0.02$), while D and AB are moderately important; all other included terms have large p -values. The corresponding estimated effects are approximately

$$A \approx -10, \quad D \approx 5, \quad AB \approx 4.5,$$

with the remaining main effects and interactions within about ± 2 .

Residual diagnostics (residuals versus fitted and a normal Q-Q plot) show no serious departures from model assumptions, although only three error degrees of freedom limit the power of these checks. Overall, the four-block design confirms that A is the dominant factor, with D and AB also influential, while the confounded interactions ABD , ABC , and CD are absorbed into the block effects.

7.7 Using the data from the 2^5 design in Problem 6.30, construct and analyze a design in two blocks with $ABCDE$ confounded with blocks.

Solution: Problem 6.30 is an unreplicated 2^5 design in coded factors

$$A, B, C, D, E \in \{-1, +1\},$$

representing aperture setting, exposure time, development time, mask dimension, and etch time. To create two blocks of 16 runs each with $ABCDE$ confounded, we compute

$$ABCDE = ABCDE$$

and assign runs with $ABCDE = -1$ to block B1 and runs with $ABCDE = +1$ to block B2. Thus

$$\text{block} \equiv ABCDE.$$

We fit a linear model including block, all main effects, and selected two-factor interactions ($AB, AC, AD, AE, BC, BD, BE, CD, CE, DE$). The ANOVA shows that block (and thus $ABCDE$) has negligible effect ($p \approx 0.75$). In contrast, factors A , B , and C and the interaction AB are highly significant (all $p \ll 0.01$); all other terms have small sums of squares and large p -values.

From the fitted coefficients, the main estimated effects (twice each coefficient) are approximately

$$A \approx 11.8, \quad B \approx 33.9, \quad C \approx 9.7, \quad AB \approx 7.9,$$

with all other effects within about ± 1.2 . Using only these important terms, a reduced regression model in coded units is

$$\hat{y} = 30.63 + 5.91A + 16.97B + 4.84C + 3.97AB,$$

which explains essentially all of the variation in yield ($R^2 \approx 0.997$). Residual plots and a Shapiro–Wilk test ($p \approx 0.42$) indicate an adequate fit.

Thus, confounding $ABCDE$ with blocks does not change the substantive conclusions from the original 2^5 design: yield increases when A and C are at their high levels and when B is at its high level, with an additional positive AB interaction. Factors D and E have relatively small effects.

8.4 Problem 6.30 describes a process improvement study in the manufacturing process of an integrated circuit. Suppose that only eight runs could be made in this process. Set up an appropriate 2^{5-2} design and find the alias structure. Use the appropriate observations from Problem 6.28 as the observations in this design and estimate the factor effects. What conclusions can you draw?

Solution: With five two-level factors and only eight runs, we use a $2^{5-2} = 2^3$ fractional factorial design. Take A, B, C as basic factors and define

$$D = AB, \quad E = AC.$$

Then the defining relation is obtained from

$$I = ABD, \quad I = ACE, \quad (ABD)(ACE) = A^2BCDE = BCDE,$$

so

$$I = ABD = ACE = BCDE.$$

This design is resolution III: each main effect is aliased with certain two-factor interactions.

The eight-run 2^3 design and responses come from Problem 6.28 (Table P6.6), using the run means as the single response for each (A, B, C) combination. Generating D and

E via the relations above and fitting a saturated model in the columns A, B, C, D, E yields the following large effect estimates (twice each coefficient):

$$E \approx 5.0, \quad D \approx 3.3, \quad C \approx 3.0, \quad BC \approx 3.5,$$

while A and B have much smaller effects (about -1.8 and 0.8).

Because of aliasing, each of these large effects represents a bundle of effects; for example,

$$E \equiv AC \equiv BCD, \quad D \equiv AB \equiv BCE.$$

Thus the eight-run fraction tells us that the alias groups associated with C , D , and E (and their interactions with A and B) are important, whereas A and B appear relatively unimportant. A follow-up design, such as a fold-over, would be required to de-alias these effects and determine which specific main effects and interactions are truly active.